make up the digital representation of the house. This includes the structure of the house and the relationships between rooms, appliances, sensors, and documents such as manuals, service records and photographs. As far as possible I have followed the RealEstateCore ontology as defined in DTDL although I have had to extend it in parts where the ontology did not define the relationships I need. Figure 5.9 shows a visualization of my complete home knowledge graph as shown in Memgraph Lab, and the detail of a small fragment showing node and edge types. Details of how to load and visualize this graph are provided in GitHub.

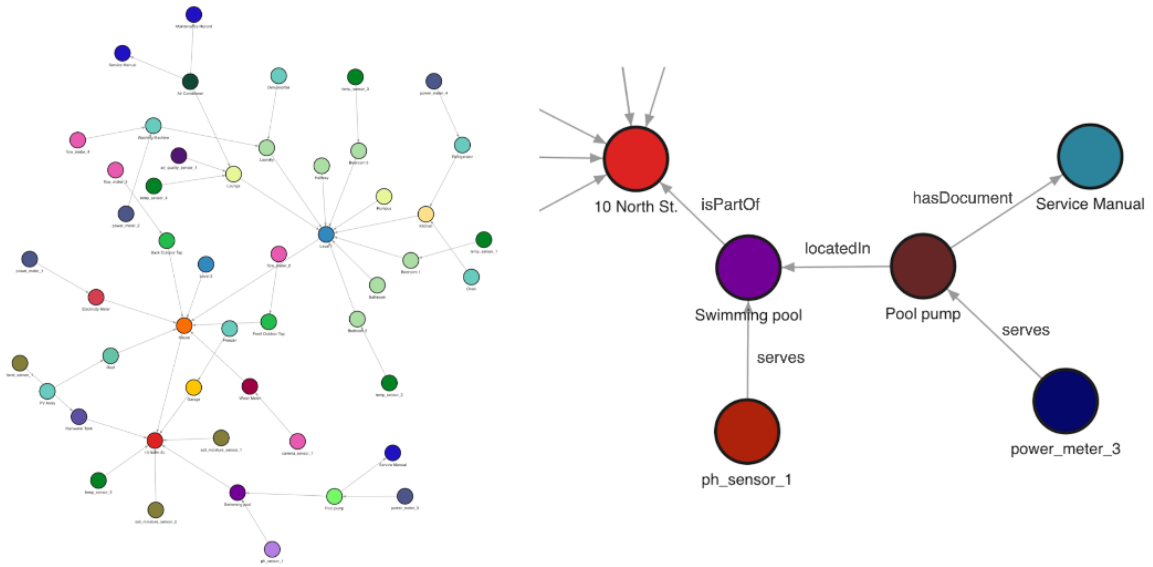


Figure 5.9 The complete knowledge graph of my home shown at left, with the detail of the relationships between the swimming pool, pump, related sensors and documents shown at right.

One of the benefits of modelling my home as a knowledge graph in this way is that I continue to build the system, discovering new entities that I want to model, and different relationship types, I can easily add them to the graph without complex schema changes.

NOTE

Willow (https://willowinc.com/) uses this same approach of a knowledge graph based on DTDL to create a digital twin of the iconic One Manhattan West skyscraper in New York city.